

SIMTAC (Simulation and Integrated Medical Training Advancement Centre)

Scenario Development Worksheet

1. Course and Trainee Information

Course title	L13 Resuscitation Drill
Department	L13
Scenario	2
Target learner group	Enrolled Nurses, Registered Nurses
No of trainees for this scenario	4
No of instructors for this scenario	2
Level of experience	Novice / Intermediate / Advanced / Expert
Pre-requisite knowledge	BCLS / LSCN trained

2. Specific Learning Objectives

In this scenario, the team is expected to:

1	Recognise early signs of deterioration.
2	Correct escalation to the medical team on patient's condition.
3	Coordinate with each other to perform an effective resuscitation.
4	Perform resuscitation skills with confidence.

3. Clinical / Environment Setting

Setting required	Remarks
General Ward	

4. Instructor Information

1	L13 ANCs / NCs / NEs
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5. Confederate Information

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6. Trainees Role

1	Nurse 1: Discover patient / AED & Drugs Nurse 2: CPR Nurse 3: Airway Nurse 4: Runner
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7. Patient Information

Name:	Boh Lub Dub		
ID No. / NRIC	S1234567A	Race:	Chinese
Age:	68	Gender:	Male
Weight:	-	Height:	-
Past medical and social history	Asthma Type 2 Diabetes Mellitus (DM) Hypertension Congestive heart failure		
Current medications	PO Aspirin 100mg OM PO Bisoprolol 2.5mg SL Glyceryl Trinitrate (GTN) 0.5mg		
Food/drug allergies	NKDA		

☐ Show information on monitor before starting scenario

☐ Make information available during the simulation

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Primary care provider	Dr Bay Xiao Xin	
History of presenting complaint	Shortness of breath Chest pain, radiating to the left shoulder	

8. Scenario Information

Time allocated for scenario	15min
Time allocated for debrief	15min
Scenario summary	Mr Boh Lub Dub, is a 68-year-old Chinese male, admitted for shortness of breath, and left-sided chest pain, radiating to the left shoulder. He is diagnosed with NSTEMI and is now in the general ward for monitoring. On the second day of hospitalisation, Mr Boh suddenly reports severe chest pain radiating to his left arm and jaw. He was diaphoretic, pale, and appears to be in distress. He is now calling for assistance. Please attend to him.
Scenario information to trainees	Mr Boh Lub Dub, is a 68-year-old Chinese male, admitted for shortness of breath, and left-sided chest pain. He is diagnosed with NSTEMI and is now in the general ward for monitoring. Past medical history: - Asthma - Type 2 Diabetes Mellitus (DM) - Hypertension - Congestive heart failure Medications ordered: - PO Aspirin 100mg OM - PO Bisoprolol 2.5mg OM - SL Glyceryl Trinitrate (GTN) 0.5mg PRN On the second day of hospitalisation, at 0800hrs, Mr Boh suddenly reports severe chest pain radiating to his left arm and jaw. He appears in pale and in distress. He is now calling for assistance. Please attend to him.
To move to the next state	☐ Next ☐ Auto ☐ Handler

9. Scenario Flow

State	Vital signs	Physical exam	Inv	Expected actions	Remarks	Instructor Control
NSTEMI	BP: 96/55mmHg PR: 120 bpm RR: 24 bpm SPO2: 93%	- Shortness of breath - Diaphoretic - Left sided chest pain – described		1st rescuer: - Administer SL GTN 0.5mg as ordered in EMR - Perform 12-lead ECG - Inform team doctor using SBAR		- Click on cycle BP and ECG when participant puts on BP

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	Rhythm: Sinus tachycardia with ST elevation	as "sharp" and "spreading to left arm and jaw"				cuff
VT	BP: - PR: - RR: - SPO2: - Rhythm: VT			1st rescuer: - Check for danger and assess for responsiveness. - Activate code blue at bedside. - Check for pulse and breathing. - Verbalise to 2nd rescuer to call 1599 to activate code blue team, call primary team, and get E-trolley with AED. - Perform one-rescuer CPR. 2nd rescuer: - Activate code blue team, call primary team, and get E-trolley. - Say "E-trolley and AED is here." 1st rescuer: - To verbalise to 2nd rescuer to take over CPR. - To verbalise to 3rd rescuer to provide ventilation via BVM. - Turn on defibrillator to 'AED' mode, connect therapy cable, and apply multifunction pads on patient. - Remove clothing and prepare the following where applicable: • Dry the chest, if wet or sweaty. • Shave the chest, if needed. • Shift any jewellery away from defibrillation pad. • Place defibrillation pads 4 fingers away from pacemaker. • Remove medication patches. - Ensure CPR is ongoing while applying pads on patient to minimise interruptions to CPR. - Place 1st pad at the upper right sternal border, just below the right clavicle. - Place 2nd pad below the left nipple at the mid axillary line. - Smooth down both pads firmly on the chest to minimise air pockets, which may cause burns and ineffective defibrillation. 3rd rescuer: - To determine appropriate size of oral airway and provide	If 2 nd rescuer is unsure how to activate code blue team and call primary team, 1 st rescuer to verbalise for 2 nd rescuer to take over CPR.	Voice of angel: - 2nd and 3rd rescuer may come in.

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				ventilation via BVM (30:2). 1st rescuer: - When AED prompts, "Analysing patient", instruct 2nd and 3rd rescuer to stop CPR. - When AED prompts, "Shock advised", perform pulse check and ensure pulse absent before delivering shock. - Say "Stand clear" and perform visual check to ensure no one is touching the patient and oxygen source is pointing away from the patient. - Confirm rhythm on the monitor and press SHOCK.		
VF	BP: - PR: - RR: - SPO2: - Rhythm: VF			1st rescuer: - Instruct 2nd and 3rd rescuer to perform CPR and ventilation for 2 min and follow voice prompt. - Prepare emergency drugs - Handover to Dr: • Time of collapse • Number of shocks given • Any medical history and medications - Ask if Dr would like to continue resuscitation with AED / manual mode - Ask if Dr would like to administer: • IV Adrenaline (1:10000) • IV Amiodarone 300mg - After 2 min, when AED prompts, "Analysing patient", instruct 2nd and 3rd rescuer to stop CPR. - When AED prompts, "Shock advised", Say "Stand clear" and perform visual check to ensure no one is touching the patient and oxygen source is pointing away from the patient. - Confirm rhythm on the monitor and press SHOCK.	1 x Co-facilitator to be assigned the doctor role.	Voice of angel: - The code blue team doctor has arrived. - 4th rescuer may come in. Co-facilitator (Doctor): - Ensure adequate handover - Administer appropriate drugs - Perform intubation (Prompt accordingly & take note of areas done well and areas for improvement to be addressed during debrief)
PEA	BP: - PR: - RR: - SPO2: -			1st rescuer: - Instruct 2nd and 3rd rescuer to perform CPR and ventilation for 2 min and	If 4 th rescuer does not know how	

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	Rhythm: Sinus rhythm			follow voice prompt. - After 2 min, when AED prompts, "Analysing patient", instruct 2nd and 3rd rescuer to stop CPR. - When AED prompts, "Shock NOT advised", check for pulse. - Pulse is absent, instruct 2nd and 3rd rescuer to continue CPR and ventilation. - Delegate 4th rescuer to prepare intubation set and suctioning apparatus if not already prepared. 4th rescuer: - Assist Dr in performing intubation.	to prepare the intubation set, to re-delegate and instruct 4 th rescuer to take over.	
Sinus rhythm	BP: 80/60mmHg PR: 88 RR: 12 SPO2: 95% Rhythm: Sinus rhythm			1st rescuer: - After 2 min, when AED prompts, "Analysing patient", instruct 2nd and 3rd rescuer to stop CPR. - When AED prompts, "Shock NOT advised", check for pulse. - Pulse is present, patient is intubated. - Instruct 2nd rescuer to stop CPR, and 3rd rescuer to continue manual ventilation 12 breaths per minute until mechanical ventilator is connected. - Verbalise immediate post-resuscitation management. • Blood investigations: CBG, ABG, FBC, Renal Panel, PT/PTT • 12-lead ECG • Chest X-ray • Obtain standby ventilator from ward 6A • Ensure ICU bed is secured and prepare for transfer to ICU • Inform family		

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10. Equipment			
Category	Item	Quantity	Remarks
Airway and breathing	Intubation set	1	
	Venti mask	1	
	Non-rebreather mask	1	
Intravenous fluids	Normal saline 0.9%	1	
	IV drip set	1	
	IV burette set	1	
Medications	E-Drug kit containing Adrenaline, Amiodarone, Lignocaine	1	
Personal			
Monitoring	Vital signs monitor		
	12-lead ECG machine		
Diagnostic			
Investigations			
Moulage			
Other equipment	E-trolley with Zoll defibrillator	1	
	Phone	1	

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11. Debrief Information

No of debriefers	8 participants + 2 facilitators
Debrief room set up, if required	Chairs
Debrief description	- Clarify roles in a resuscitation, emergency drugs dosage and dilution, resuscitation algorithm. - Address areas done well and areas for improvement - Summarise key learning points
Logistics required e.g markers, flipchart	Whiteboard markers & Whiteboard

12. Simulator / Standardised Patient / Task Trainer Preparation

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13. Patient Monitor Setup

Monitor Layout	☐ 5 waveform ☐ 4 waveform ☐ 3 waveform ☐ Big Num layout	
Parameters	☐ Primary Electrocardiogram ☐ Secondary Electrocardiogram ☐ ABP - Arterial Blood Pressure ☐ SpO₂ (Plethysmogram) – Oxygen saturation ☐ PAP - Pulmonary Artery Pressure ☐ CO ₂ – End-tidal Carbon Dioxide ☐ pH – Atrial pH ☐ CVP - Central Venous Pressure ☐ NBP- Non-invasive Blood Pressure ☐ TOF - Train Of Four	☐ N ₂ O - Nitrous Oxide ☐ O ₂ - Oxygen pressure ☐ Pulse ☐ WP – Wedge Pattern ☐ RR - Airway Respiration Rate ☐ C.O. - Cardiac Output ☐ TPeri - Peripheral Temperature ☐ Tblood - Invasive blood temperature ☐ ICP - Intracranial pressure ☐ AGT – Agent ☐ HR- Heart Rate

Note: A maximum of 5 waveforms and 14 numeric parameters can be shown at the same time. Both waveform and numeric are displayed together.

14. Document Information

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